

JEE MAIN REVISION MASTER: SALT ANALYSIS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Salt Analysis — Core Principles & Governing Laws	4 PYQs	Core Principle
Standard	Salt Analysis — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis
Standard	Salt Analysis — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Salt Analysis — Core Principles & Governing Laws	NH_4OH , H_2SO_4 , NH_3

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Salt Analysis — Core Principles & Governing Laws (Moderate Yield — 4 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Salt Analysis.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Salt Analysis.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Salt Analysis PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "When salt BA is treated with Conc. H_2SO_4 reddish brown gas is liberated. The aqueous solution of BA gives pale yellow ..." [Ans: D]
- [2025 Shift] Q#2 (MCQ): "The group reagent $\text{NH}_4\text{Cl(s)}$ and aqueous NH_3 will precipitate which of the following ion?..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The sodium fusion extract is boiled with concentrated nitric acid while testing for halogens. By doing so, it..." [Ans: N/A]

3.2 Salt Analysis — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Salt Analysis.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "When salt BA is treated with Conc. H_2SO_4 reddish brown gas is liberated. The aqueous solution of BA gives pale yellow ..." **[Ans: D]**
- **[2025 Shift] Q#2 (MCQ):** "The group reagent $NH_4Cl(s)$ and aqueous NH_3 will precipitate which of the following ion?..." **[Ans: N/A]**

3.3 Salt Analysis — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Salt Analysis.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "When salt BA is treated with Conc. H_2SO_4 reddish brown gas is liberated. The aqueous solution of BA gives pale yellow ..." **[Ans: D]**
- **[2025 Shift] Q#2 (MCQ):** "The group reagent $NH_4Cl(s)$ and aqueous NH_3 will precipitate which of the following ion?..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

When salt BA is treated with Conc. H_2SO_4 reddish brown gas is liberated. The aqueous solution of BA gives pale yellow precipitate with $AgNO_3$ solution. Which of the following anion (A^-) is present in the salt BA?

- (A) Cl^-
- (B) CO_3^{2-}
- (C) SO_4^{2-}
- (D) Br^- **[Correct]**

Correct Answer: D

Solution Analysis: The anion present is Br^- . When a bromide salt is treated with concentrated H_2SO_4 , reddish brown vapours of bromine are liberated. First, hydrogen bromide is formed: $BBr + H_2SO_4 \rightarrow BHSO_4 + HBr$. Then HBr reduces concentrated H_2SO_4 and gets oxidised to bromine: $2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$. The bromine vapours, Br_2 , are reddish brown in colour. Also, the aqueous solution of the salt gives a pale yellow precipitate with $AgNO_3$.